



Product information

Product full identity:

Polycarbonate

PC has excellent optical properties and a high gloss surface. Easy to fabricate PC shows exceptional performance with a wide operating temperature range (-40°C to +135°C). Other important benefits are high mechanical, thermal and electrical properties also virtually unbreakable in normal use. Also available in UV grade.

Properties

- » 25% more impact resistant than PETG
- » Virtually unbreakable
- » Good fire properties & self extinguishing
- » Suitable for thermoforming

Applications

- » Machine guards
- » Riot shields
- » Visors
- » Signs
- » Windows

This document contains

- » Technical Datasheet (Page 2)
- » Chemical Datasheet (Page 3)
- » Safety Datasheet (Pages 4-6)

For any further information regarding food, fire and water certificates then please contact the sales team on 0114 256 0889

Technical Properties

Physical Properties	Test	Unit	Result
1. Specific gravity	ISO 1183	g/cm ³	1.2
2. Water absorption	ISO 62	%	0.35
3. Maximum service temp. Upper temp limit (no stronger mechanical stress involved)	-	°C	120
Lower temp limit	-	°C	-30
Mechanical Properties	Test	Unit	Result
1. Tensile strength at yield	ISO 527	MPa	60
2. Elongation at yield	ISO 527	%	6
3. Tensile strength at break	ISO 527	MPa	72
4. Elongation at break	ISO 527	%	150
5. Impact strength	ISO 179	kJ/m ²	no break
6. Notch impact strength	ISO 179	kJ/m ²	55
7. Ball indentation / Rockwell hardness	ISO 2039-1	MPa	M72 / R118
8. Shore-D	DIN 53505	-	-
9. Flexural strength	ISO 178	MPa	97
10. Modulus of elasticity	ISO 527	MPa	2300
Thermal Properties	Test Method	Unit	Result
1. Vicat-softening point VST/B/50	ISO 306	°C	151
2. Heat deflection temperature HDT/B	ISO 75	°C	143
HDT/A	-	°C	151
3. Coefficient of linear thermal expansion	ISO 75-2	(x10-5/°C)	0.68
4. Thermal conductivity at 20 °C	DIN 8302	W/(m*K)	0.20
Electrical Properties	Test Method	Unit	Result
1. Volume resistivity	VDE 0303	Ω x m	3x10 ¹⁴
2. Surface resistivity	-	Ω	6x10 ¹⁶
3. Dielectric constant at 1MHz	-	-	2.7
4. Dielectric loss factor at 1 MHz	DIN 53483	-	0.01
5. Dielectric strength	VDE 0303	kV/mm	17
6. Tracking resistance (CTI)	IEC 60112	-	250
Additional Data	Test Method	Unit	Result
1. Bondability	-	-	+
2. Food compliance	FDA	-	-
3. Flammability	UL 94	-	HB

All The above information is for guide purposes only. The data has been taken from standard test results provided by our manufacturers.

Key:

Yes	Limited	No data
+	0	-

Chemical Properties

Agent	Conc %	Working	Temp	Agent	Conc %	Working	Temp
		20°C	60°C				
Acetic Acid	100	-	-	Hydrofluoric acid	40	-	-
Acetone	100	-	-	Hydrogen peroxide	10	+	
Ammonia	Conc.	-	-	Hydrogen Sulphide		+	
Ammonium chloride		+	+	Isopropyl Alcohol	100	o	
Amyl Alcohol				Mercurochrome		o	
Benzene		-	-	Methyl alcohol	100	-	-
Bleaching Solution	12,5 Cl	-	-	Methyl ethyl ketone	100	-	-
Boric Acid	100	+		Methylene chloride	100	-	-
Brake Fluid				Nitric acid	10	o	
Butyl Acetate		-	-	Nitric acid	50	-	-
Calcium Chloride		+	+	Nitrobenzine		-	-
Carbon disulphide	100	-	-	Oxalic Acid		+	
Carbon Tetrachloride		-	-	Ozone, gas	ca. 0,5 ppm	o	
Chlorine, gas	100	o		Paraffin Oil	100	+	
Chlorobenzene	100	-	-	Perchlorethylene		-	-
Chloroform		-	-	Petroleum	100	o	
Citric Acid	10	+	+	Petroleum, aromatic free	100	o	-
Cresol		-	-	Phenol, aqu	ca.9	-	-
Cyclohexanone	100	-	-	Phosphoric Acid	50	+	
Cyclohexene	100	-	-	Potassium hydroxide liquor	50	-	-
Diesel Fuel		o		Propyl alcohol		+	
Ethyl acetate	100	-	-	Pyridine		-	-
Ethyl alcohol	96	+/-	o	Silicone oil		+	+
Ethylene Chloride	100	-	-	Sodium carbonate. aqu		+	+
Formic Acid	10	+		Sodium chloride, aqu		+	+
Frost protection agent		+		Sodium Hydroxide liquor	60	-	-
Fuel, aromatic free		-		Sodium hydrogen sulphite		+	
Glycerine	100	+/-	o	Sodium nitrate, aqu		+	+
Glycol	100	+		Sodium thiosulfate			
Heating oil		o		Sulphuric Acid	96	-	-
Heptane	100	-	-	Tetrahydrofuran	100	-	-
Hydrochloric acid	10	+		Toluene	100	-	-
Hydrochloric acid	conc.	o/-	-	Trichlorethylene	100	-	-
				Xylene		-	-

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Key:

Yes	Limited	No data
+	o	-

Safety Properties

Substance / preparation and company detail

Polycarbonate (PC)
Direct Plastics Limited
Rother Valley Way,
Holbrook,
Sheffield,
S20 3RW
0114 2560889

Composition / indications to components

Chemical characterization : Polycarbonate (PC)

Hazardous substances : Product contains no hazardous ingredients liable to be disclosed.

Possible dangers

Classification : Not classified

Physical/ chemical hazards : Flammable

Health risks : Dust can cause mechanical irritation.

Hazards for the environment: Based on our information, there is no danger to the environment.

The product is according to Directive 1999/45/EC and its annexes are not classified as dangerous.

First-aid measures

General information : The product is being classified as non-toxic.

In case of inhalation : In case the plastic burns and combustion gases are inhaled, remove person to fresh air and keep warm and get medical help if necessary.

In case of skin contact : Burns caused by molten material on skin need to be rapidly cooled down with water; do not attempt removal of plastic without medical assistance. If irritation develops, seek medical attention.

In case of eye contact : Flush eyes well with copious quantities of water. Seek medical attention, if irritation persists.

In case of ingestion : The product is non-toxic; no first aid procedures are required.

First-fighting measures

Suitable extinguisher: Water, foam, gaseous and dry extinguishing media

Particular endangerments by fire fighting and hazardous combustion products: Hazardous combustion products may emerge, apart from harmless Water (H₂O); carbon dioxide (CO₂) and mainly carbon monoxide (CO) depending on the amount of available environmental oxygen, containing ketones and aldehyde. Formation of further decomposition and oxidation products depends upon the fire conditions. Under special fire conditions traces of other toxic substances are possible.

Fire fighting: Approved pressure demand breathing apparatus and protective clothing should be used for all fires.

Additional Information: Residues after the fire, after appropriate rules dispose.

Measures in case of unintended release

Personal precautions : N/A

Environmental precaution : N/A

Methods for cleaning up : Mechanical removal

Handling and storage

Advice on safe handling: During machining of the stock shapes, evacuate swarf to prevent slipping or tripping.

Storage: Store inert product dry and cool. Keep storage and working areas sufficiently ventilated. Keep away from source of flame, heat and ignition. Due to the risk of collapsing, do not stack more than 2 pallets on to of each other. Pallets should not stack on to of each other along aisles.

Safety Properties

Limitation of exposition

Ingredients with occupational exposure limits to be monitored : none

General protective and Hygiene measures : Keep the workplace sufficiently ventilated; thereby smoking; eating and drinking are not allowed.

Continuous supply of fresh air to the workplace together with removal of processing fumes through exhaust systems is recommended. Avoid breathing in gaseous degradation products and dust that may result by material overheating.

Hand protection : Safety gloves in case of contact with warm material

Eye protection : Safety goggles or shield during machining

Body protection : Working clothes

Respiratory protection : Adequate ventilation at workplace is required

Physical and chemical characteristics

Aggregate : solid

Colour : product-specific

Odour : slight, product specific

Safety related facts

Boiling point : N/A

Melting point : 150 - 160 °C (DIN/EN/ISO 3146)

Corrosion temperature : N/A

Flash point : N/A

Self ignition temperature : 560 °C (ASTM D1929)

Explosion hazard or limit : non explosive

Oxidizing characteristics : None

Density (20 °C) : 1.20 g/cm³ (ISO 1183)

Solubility (in Water 20 °C) : insoluble

Viscosity : N/A

Additional Information : None

Stability and reactivity

Conditions to avoid: Temperatures above melting point

Material to avoid: Strong oxidant

Hazardous decomposition products: Carbon monoxide CAS-Nr. 630-08-0

Toxic information

Acute toxicity: no data existing, except those mentioned below:

Water-insoluble

Material Tests:

LD-50, oral: (male rat) >3,200 mg/kg (highest dose rate tested)

LD-50, oral: (male mouse) >3,200 mg/kg (highest dose rate tested)

Dermal LD-50: (Guinea pig) >1,000 mg/kg

Skin irritation (guinea pig) Slight irritation

Irritation of eyes (rabbit, unwashed eyes) Slight

Irritation of eyes (rabbit, washed eyes) Slight

Guinea pig None

Additional information:

In our experience and according to information available to us the product is not harmful to health provided

Safety Properties

Ecological information

The material does not harm the environment but is not biologically degradable.

Waste-disposal information

The product must be disposed in accordance with the local authorities.

Transport information

Not classified as hazardous under transport regulations.

Regulations

The product does not require a hazard warning label in accordance with EC directives.

Further information

The information is based on our current knowledge. They are meant to describe our products in respect to safety requirements. They do not represent any guarantee of the described product in the sense of the legal guarantee regulations.